AI-BASED VEHICLE DETECTION, TRACKING AND TRAFFIC VOLUME ANALYSIS USING YOLO AND BYTETRACK

A Computer Vision-Based Application in Transportation Engineering

Project Report

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Technologies Used: Python, YOLO11n, ByteTrack, OpenCV, Pandas and Matplotlib

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ABSTRACT

Traffic volume studies are an important component of transportation engineering, as they provide information about the number and types of vehicles using a road. Conventional traffic surveys often involve manual observation and vehicle counting, which can be time-consuming and may introduce human errors.

This project develops an AI-based vehicle detection, tracking and traffic analysis prototype using the YOLO11n object detection model, ByteTrack multi-object tracking algorithm and OpenCV video-processing library.

The system processes traffic video footage, identifies vehicle categories, assigns tracking IDs and records vehicle crossings at a predefined virtual counting line. The resulting event data is analysed using Python and visualised through a traffic analytics dashboard.

The prototype processed 482 frames from a traffic video approximately 16.07 seconds long and recorded 29 vehicle-crossing events. The recorded events consisted of 25 cars and 4 buses.

The project demonstrates the potential application of computer vision in automated traffic monitoring. However, the results require further manual validation before the system can be considered a reliable traffic-survey tool.

Keywords: Artificial Intelligence, Vehicle Detection, YOLO, ByteTrack, Computer Vision, Traffic Volume, Transportation Engineering.

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1. INTRODUCTION

Traffic volume is one of the fundamental parameters used in transportation engineering. It represents the number of vehicles passing a particular point on a road during a specified period.

Traffic volume information is useful for:

- Roadway planning and geometric design.
- Traffic management and control.
- Intersection improvement.
- Road capacity assessment.
- Transportation infrastructure planning.
- Vehicle composition studies.

Traditional traffic volume studies involve manually observing vehicles passing a selected point and recording their numbers and categories.

Although manual surveys are widely used, they can be time-consuming and may be affected by human observation errors.

Recent developments in Artificial Intelligence (AI) and computer vision provide opportunities to automate vehicle detection and traffic monitoring using video footage.

This project applies object detection and multi-object tracking techniques to road traffic footage to identify vehicles, track their movement and estimate traffic volume.

2. OBJECTIVES

The main objectives of this project are:

1. To detect vehicles from traffic video footage using the YOLO object detection model.
2. To classify vehicles into categories such as cars, motorcycles, buses and trucks.
3. To track detected vehicles across successive video frames using ByteTrack.
4. To count vehicles crossing a predefined virtual counting line.

5. To record vehicle IDs, categories, crossing times and directions.
6. To analyse vehicle composition and observed traffic flow.
7. To develop a visual dashboard for presenting traffic analysis results.
8. To explore the application of AI-based computer vision in transportation engineering.

3. SOFTWARE AND TECHNOLOGIES USED

The project was developed using the following software and technologies.

Technology	Application
Python	Main programming language
YOLO11n	Vehicle object detection
ByteTrack	Multi-object tracking
OpenCV	Video reading and frame processing
Pandas	Dataset preparation and analysis
Matplotlib	Data visualisation
Google Colab	Cloud-based development environment

3.1 YOLO11n

YOLO11n is a lightweight object detection model used to identify objects in individual video frames.

In this project, it was used to detect vehicles and generate bounding boxes around the detected objects.

The model also provides class predictions and confidence scores for the detected objects.

3.2 ByteTrack

ByteTrack is a multi-object tracking algorithm used to associate detected objects across successive video frames.

It assigns tracking IDs to detected objects, enabling the system to follow vehicles as they move through the camera view.

This helps distinguish individual tracked objects from repeated detections of the same object across multiple frames.

3.3 OpenCV

OpenCV is a computer vision library used for reading and processing video frames.

In this project, it was used to:

- Read the uploaded traffic video.
- Extract individual frames.
- Draw bounding boxes and virtual counting lines.
- Generate annotated video output.

3.4 Pandas and Matplotlib

Pandas was used to organise vehicle-crossing events into structured datasets.

Matplotlib was used to generate graphs and visualise vehicle composition, category-wise counts and vehicle crossing times.

4. METHODOLOGY

The project follows a video-processing pipeline consisting of vehicle detection, classification, tracking, crossing detection, data extraction and traffic analysis.

4.1 Video Input

A traffic video was uploaded into the Google Colab environment and processed frame by frame.

Video specifications:

Parameter	Value
Total processed frames	482
Frame rate	Approximately 30 FPS

Video duration	Approximately 16.07 seconds

The video was used as the input dataset for the vehicle detection and tracking system.

4.2 Vehicle Detection Using YOLO11n

The YOLO11n model was used to identify vehicles in individual video frames.

The selected vehicle classes from the COCO dataset were:

Class ID	Vehicle Category
2	Car
3	Motorcycle
5	Bus
7	Truck

The model generated bounding boxes around detected vehicles and assigned corresponding class predictions.

4.3 Multi-Object Tracking Using ByteTrack

After vehicle detection, ByteTrack was used to associate detections across consecutive frames.

Each tracked object was assigned a tracking ID.

This enabled the system to monitor vehicle movement across the video and maintain object identities between frames.

It is important to note that tracking IDs are not necessarily equal to the number of physical vehicles. A vehicle may lose its tracking ID and receive another one because of occlusion, missed detections or tracking interruptions.

4.4 Virtual Counting Line

A horizontal virtual counting line was placed at approximately 60% of the video frame height.

The system monitored vehicle positions across successive frames to identify when a tracked vehicle crossed the virtual line.

When a crossing was identified, the system recorded the vehicle as a crossing event.

The original counting implementation ensured that each tracking ID was counted only once.

4.5 Vehicle Event Dataset Generation

For each recorded crossing event, the system stored the following information:

Field	Description
Vehicle_ID	Tracking ID assigned to the vehicle
Vehicle_Type	Predicted vehicle category
Time_Seconds	Time at which the crossing was recorded
Direction	Direction of movement across the line

The information was stored in a Pandas DataFrame.

The resulting dataset was used for further traffic analysis and exported for documentation.

4.6 Data Visualisation

Matplotlib was used to develop a traffic analytics dashboard.

The dashboard presents:

- Total recorded vehicle crossings.
- Video duration.
- Observed traffic flow rate.
- Hourly-equivalent flow rate.
- Vehicle composition.
- Category-wise vehicle counts.
- Vehicle crossing timeline.

5. METHODOLOGY FLOWCHART

The overall project methodology can be represented as follows:

Step 1: Traffic Video Input

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Step 2: Frame Extraction Using OpenCV

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Step 3: Vehicle Detection Using YOLO11n

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Step 4: Vehicle Classification

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Step 5: Object Tracking Using ByteTrack

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Step 6: Virtual Counting Line

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Step 7: Vehicle Crossing Detection

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Step 8: Event Dataset Generation

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Step 9: Traffic Volume Analysis

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Step 10: Dashboard and Visualisation

6. MATHEMATICAL ANALYSIS

This section presents the mathematical calculations used to determine video duration, observed traffic flow rate, hourly-equivalent flow rate and vehicle composition.

6.1 Calculation of Video Duration

The total duration of the video is calculated using the following relationship:

Video Duration = Total Number of Frames ÷ Frame Rate

Mathematically,

T = N / F

Where:

- T = Total video duration (seconds)
- N = Total number of frames
- F = Frame rate (frames per second)

Given:

- Number of frames (N) = 482
- Frame rate (F) = 30 FPS

Substituting the values:

T = 482 / 30

T = 16.0667 seconds

Final Answer: T = 16.07 seconds

Therefore, the processed video duration is approximately 16.07 seconds.

6.2 Calculation of Observed Traffic Flow Rate

Traffic flow rate represents the number of vehicles passing a particular point per unit time.

It is calculated using:

Traffic Flow Rate = Number of Vehicle Crossings / Observation Time

Mathematically,

q = V / T

Where:

- q = Observed traffic flow rate (vehicles/second)
- V = Total number of recorded vehicle crossings
- T = Video duration (seconds)

Given:

- Total recorded vehicle crossings (V) = 29
- Video duration (T) = 482 / 30 seconds

Substituting the values:

q = 29 / (482 / 30)

q = 1.80498 vehicles/second

Final Answer: q ≈ 1.805 vehicles/second

Therefore, the observed traffic flow rate is approximately 1.805 vehicles per second.

6.3 Calculation of Hourly-Equivalent Traffic Flow Rate

The observed traffic flow rate can be converted into an hourly-equivalent value by multiplying it by the number of seconds in one hour.

Hourly-Equivalent Flow Rate = Observed Flow Rate × 3600

Mathematically,

Q = q × 3600

Where:

- Q = Hourly-equivalent flow rate (vehicles/hour)
- q = Observed traffic flow rate (vehicles/second)
- 3600 = Number of seconds in one hour

Given:

- Observed flow rate (q) = 1.80498 vehicles/second

Substituting the values:

Q = 1.80498 × 3600

Q = 6497.93 vehicles/hour

Final Answer: Q ≈ 6498 vehicles/hour

Note: This value is a mathematical extrapolation from a 16.07-second video sample. It is not a measured hourly traffic volume, peak-hour traffic volume or road capacity.

6.4 Calculation of Vehicle Composition

Vehicle composition represents the percentage contribution of each vehicle category to the total recorded traffic volume.

The percentage of a particular vehicle category is calculated using:

Vehicle Composition (%) = (Number of Crossings of the Vehicle Category / Total Vehicle Crossings) × 100

Mathematically,

$P_i = (V_i / V) \times 100$

Where:

- P_i = Percentage of a particular vehicle category (%)
- V_i = Number of recorded crossings of that vehicle category
- V = Total number of recorded vehicle crossings

A. Percentage of Cars

Given:

- Number of cars = 25
- Total recorded vehicle crossings = 29

Using the formula:

Percentage of Cars = (Number of Cars / Total Vehicle Crossings) × 100

Substituting the values:

P_car = (25 / 29) × 100

P_car = 86.2069%

Final Answer: Percentage of Cars = 86.21%

Therefore, cars constitute approximately 86.21% of the recorded vehicle crossings.

B. Percentage of Buses

Given:

- Number of buses = 4
- Total recorded vehicle crossings = 29

Using the formula:

Percentage of Buses = (Number of Buses / Total Vehicle Crossings) × 100

Substituting the values:

P_bus = (4 / 29) × 100

P_bus = 13.7931%

Final Answer: Percentage of Buses = 13.79%

Therefore, buses constitute approximately 13.79% of the recorded vehicle crossings.

C. Verification of Vehicle Composition

The sum of the percentages of all recorded vehicle categories should equal 100%.

Total Vehicle Composition = Percentage of Cars + Percentage of Buses

Substituting the calculated values:

Total Vehicle Composition = 86.21 + 13.79

Final Answer: Total Vehicle Composition = 100%

Hence, the calculated vehicle composition is consistent with the 29 recorded crossing events.

7. RESULTS AND OBSERVATIONS

7.1 Vehicle Crossing Results

The original AI crossing counter recorded the following results:

Vehicle Category	AI-Recorded Crossings
Cars	25
Buses	4
Motorcycles	0
Trucks	0
Total	**29**

7.2 Traffic Analysis Results

Parameter	Result
Video duration	16.07 seconds
Processed frames	482
Total recorded crossings	29
Observed flow rate	1.805 vehicles/second
Hourly-equivalent flow rate	Approximately 6,498 vehicles/hour

7.3 Vehicle Classification Observations

The initial crossing counter classified the 29 recorded events as 25 cars and 4 buses.

Separate frame-level detection analysis showed that YOLO also detected motorcycles and trucks in individual frames.

However, frame-level detections do not represent unique vehicles or crossing events.

The difference between frame-level classifications and crossing-event classifications indicates that vehicle classification and tracking require additional validation.

7.4 Tracking Observations

The tracking experiments produced multiple tracking IDs, including short-duration tracks.

These IDs cannot be treated as unique physical vehicle counts.

Short-duration tracks may result from:

- Vehicles entering or leaving the camera view.
- Temporary vehicle occlusion.
- Missed detections.
- Inconsistent object tracking.

The results indicate the importance of improving tracking consistency and validating the final crossing counts.

8. LIMITATIONS

The current prototype has several limitations:

1. **Short video duration:** The analysis uses approximately 16.07 seconds of footage, which is insufficient for reliable long-duration traffic studies.
2. **Classification uncertainty:** Some vehicles may be misclassified or missed by the detection model.
3. **Tracking fragmentation:** A physical vehicle may receive multiple tracking IDs if tracking is interrupted.
4. **Counting uncertainty:** The crossing count has not yet been independently verified against a manually labelled event dataset.
5. **Limited traffic analysis:** Peak-hour traffic, hourly variation and congestion cannot be reliably determined from this short sample.
6. **Camera dependency:** Detection performance can vary with camera angle, lighting conditions, vehicle overlap and image quality.

9. FUTURE SCOPE

The project can be improved through the following developments:

1. Manual ground-truth annotation and quantitative accuracy evaluation.
2. Improved vehicle detection and tracking configuration.
3. Longer-duration traffic video analysis.
4. Separate validation of cars, motorcycles, buses and trucks.
5. Direction-wise traffic volume estimation.
6. Passenger Car Unit (PCU)-based traffic analysis using appropriate reference factors.
7. Traffic density and congestion estimation.
8. Development of a real-time traffic monitoring interface.
9. Testing the system on multiple road types and camera positions.

These improvements can help transform the current prototype into a more reliable automated traffic monitoring system.

10. CONCLUSION

This project demonstrates a prototype for automated vehicle detection, tracking and traffic volume analysis using computer vision techniques.

YOLO11n was used for vehicle detection and classification, while ByteTrack was used to associate detected objects across video frames. OpenCV supported video processing, and Python-based data analysis was used to generate crossing-event records and a visual traffic analytics dashboard.

The prototype processed 482 frames from approximately 16.07 seconds of traffic footage and recorded 29 vehicle-crossing events, comprising 25 cars and 4 buses.

The observed traffic flow rate was approximately 1.805 vehicles per second, while the hourly-equivalent flow rate was approximately 6,498 vehicles per hour. The latter represents an extrapolated value rather than a measured hourly traffic volume.

The project demonstrates how AI-based computer vision can support transportation engineering applications, particularly automated vehicle monitoring and traffic data collection.

Further work involving manual validation, longer video samples and improved tracking reliability is necessary before the system can be used for dependable traffic-volume surveys.

11. PROJECT DELIVERABLES

The following outputs were generated during the project:

Deliverable	Description
Original traffic video	Input video used for analysis
Annotated traffic video	Video with vehicle detections, labels and tracking IDs
Vehicle event dataset	CSV/Excel records of crossing events
Tracking analysis	Tracking ID and duration analysis
Traffic analytics dashboard	Visual representation of traffic results
Project report	Methodology, calculations, results and limitations

12. TECHNOLOGIES AND SKILLS DEMONSTRATED

Programming and Data Analysis: Python, Pandas, Matplotlib

Artificial Intelligence: YOLO11n, Object Detection, Vehicle Classification

Computer Vision: OpenCV, Video Processing

Object Tracking: ByteTrack, Multi-Object Tracking

Civil Engineering Application: Transportation Engineering, Traffic Volume Analysis, Vehicle Composition and Traffic Monitoring

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PROJECT STATUS: WORKING PROTOTYPE COMPLETED.